

Problem Statement

The axial fan is designed to operate in an aero engine operating at an altitude of 10km with a cruise Mach number of 0.78. With a design mass flow rate of 80 [kg/s] and rotational speed of 5000 [rpm] a total-to-total pressure in the range of 1.4-1.7 is desired. This poster illustrates the design of the fan along with the procedure and analysis that lead to recommendations on implementing it for flight.

Design procedure

With the design specifications identified, an isentropic meanline analysis provides a first estimate of the duty coefficients needed to achieve the target total pressure ratio, rotor tip Mach number, meridional velocity, and hub-to-tip ratio at the specified RPM. An initial mean radius is also determined. Since the flow enters axially, the degree of reaction directly follows from the flow and work coefficients (φ and ψ), allowing definition of the flow angles.

Solidity is minimized while satisfying the Howell criterion for both rotor and stator. A minimum chord length is assumed for structural integrity, from which the pitch and blade count are derived. The enthalpy change and an initial estimate of the loss coefficients and efficiency are also calculated.

Knowing all flow angles, the incidence and deviation angles are determined. With these parameters, a CFD simulation provides a more realistic efficiency estimate, full geometry, and flow regime. The inputs can then be refined to optimize efficiency and weight.

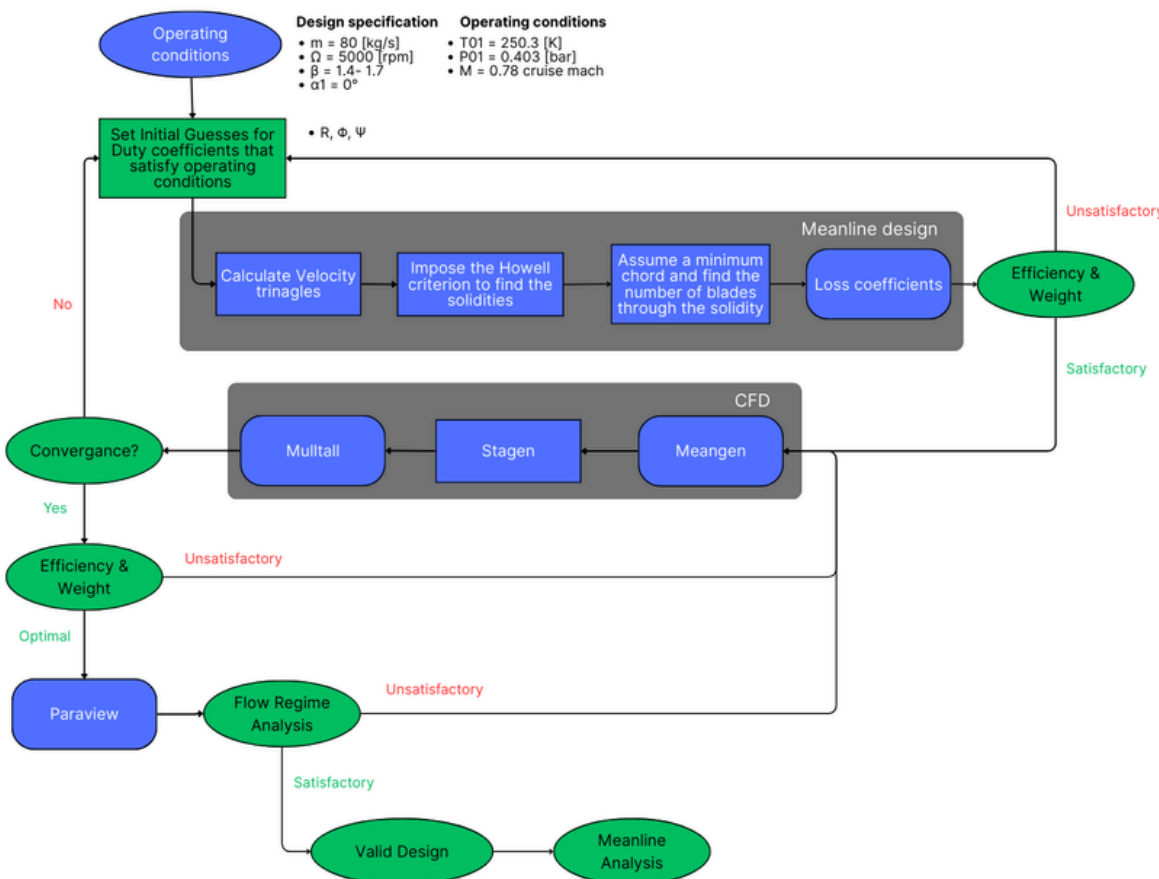


Figure 1. Flowchart of the design procedure

With the duty coefficients defined, several degrees of freedom remain in the design, including blade thickness, maximum thickness position, rotor-stator spacing, blade twist, taper ratio, and details like trailing edge thickness. Some of these parameters are varied to improve the design, although not all, to limit computational effort. Additionally, the duty coefficients themselves are adjusted to maximize efficiency while satisfying the constraints. This is expected, as flow incidence and deviation prevent achieving the initially calculated velocity triangles, leading to different optimal duty coefficients.

The meanline analysis presented here should theoretically lead to feasible design, but since the conditions in the determination of the meanline design parameters are all pass/fail and the choice of working procedure is based on engineering sense rather than measurable performance, an optimisation of the duty coefficients is desired. To avoid the arbitrary choice of any of the factors to be minimised, a pareto front optimisation is recommended, though a single or small range of points does have to be chosen to move forward due to the computation time of the CFD results.

Meanline Analysis

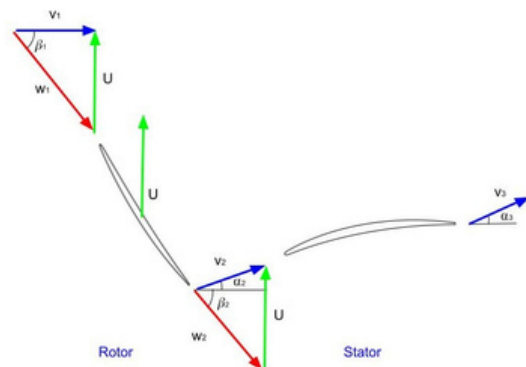


Figure 2. Velocity triangles of the fan

Duty Coefficients	
Flow coefficient	$\varphi = 0.723$
Work coefficient	$\psi = 0.571$
Degree of reaction	$R = 0.715$

Table 1. Meanline Duty Coefficients

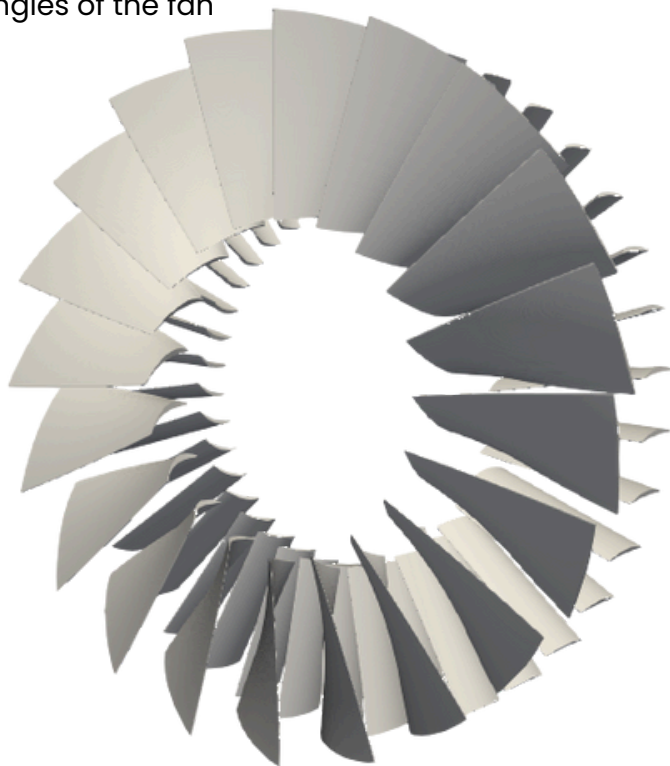


Figure 3. 3D view of the resulting fan

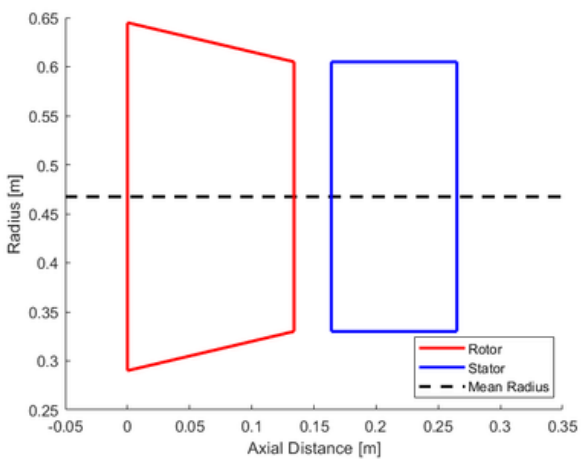
To obtain an initial estimate of the fan's efficiency, the meanline analysis accounted for blade profile losses using the Leiblein model and for Endwall losses using the Howell model. Shock losses have been ignored because the inlet Mach number is insufficiently high.

Inlet properties	Rotor	Stator
Blades	20	28
Total temperature [K]	250.3	279.2
Static temperature [K]	240.0	267.1
Total Pressure [kPa]	39.0	44.0
Static Pressure [kPa]	34.8	37.3
Loss coefficient	0.0279	0.0303
Efficiency [%]	93.97	

Table 2. Meanline Flow Properties

Design Parameters	
Mean radius	0.481
Rotor Chord	0.135 [m]
Stator chord	0.101 [m]
Enthalpy change	2348 [kW]

Table 3. Meanline Meridional Design



Forces	Lift [N]	Drag [N]	Centrifugal [N]
Rotor	2257.52	376.38	413855.71
Stator	880.92	265.04	0

Table 4. Blade Aerodynamic losses

CFD Analysis

Once a converged solution has been reached, the resulting fluid volume is analysed to provide more insight into 3D losses that have been neglected in the mean line analysis. The Mach counter plot shows that the flow through the fan blades is well-established, smooth, and continuous, indicating a started-flow regime. This is preferable as the fan is less likely to stall. To further improve the design, the row gap, the position of the maximum thickness, and the twist have been investigated to see which combination yields optimal results.

Global Parameters	
Mass flow rate	81.9 [kg/s]
Total-to-Total Efficiency	90.4 [%]
Total-to-Static Efficiency	44.1 [%]
Power required	2348 [kW]
Pressure ratio	1.41

Table 5. CFD Flow Properties

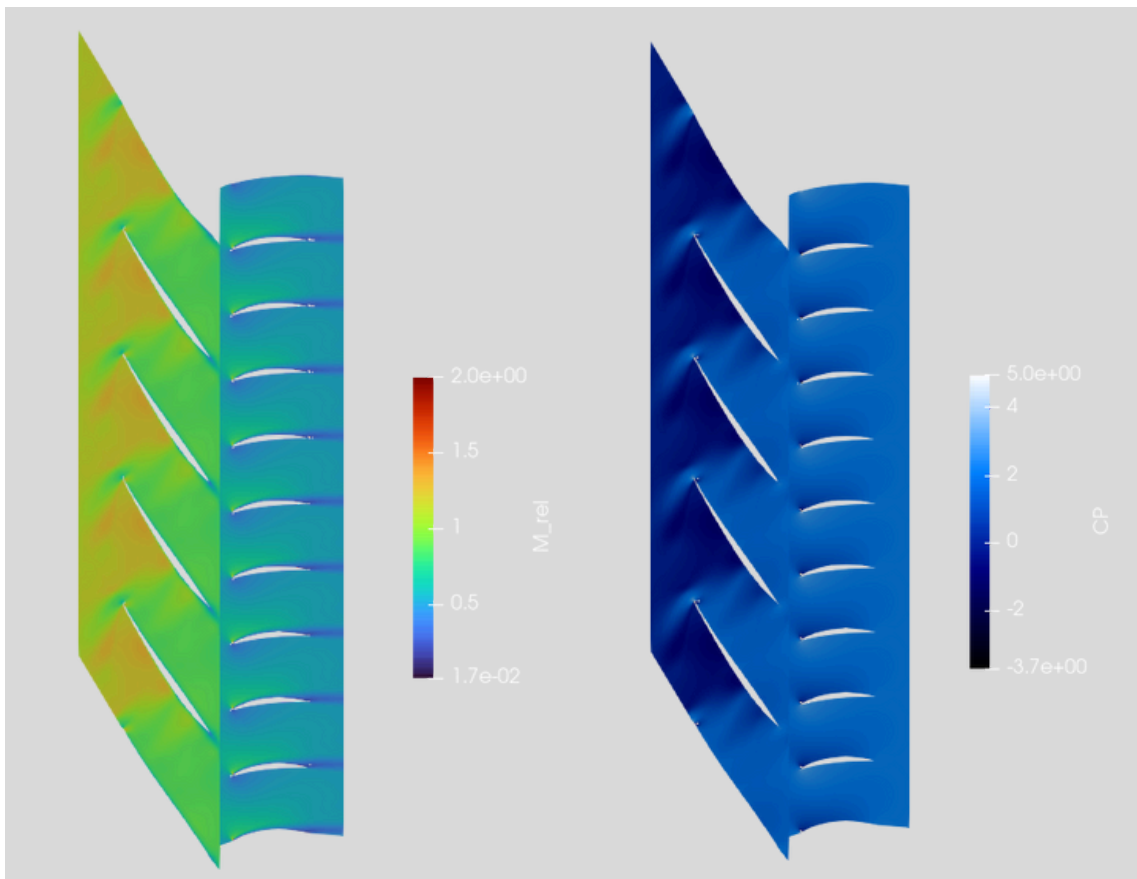


Figure 4. CFD results of the flow at the meanline

Losses at mid	
Rotor	0.117
Stator	0.128

Figure 5. Losses from CFD

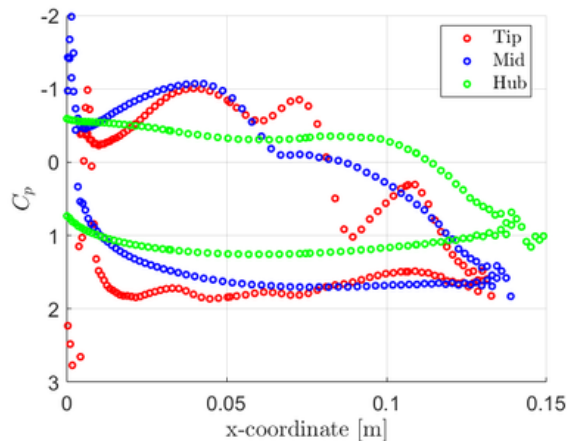


Figure 6. Dimensionless pressure plot

Conclusion And recommendations

The design procedure presented here demonstrates an axial fan for aero engine applications. While achieving an impressive 90.4% efficiency, several key improvements are recommended to enhance performance further. First, blade taper angles should be incorporated into numerical optimization to avoid unnecessarily constraining the design space. Additionally, reducing blade twist would mitigate tip leakage losses and hub flow deviation. For optimal fluid dynamic performance, blade geometry should vary along the span, providing better control over machine weight and enabling its inclusion in the optimization process. A recalculation of incidence and deviation angles is essential, as current inaccuracies reduce confidence in the CFD efficiency results. Finally, implementing a Pareto front optimization would comprehensively address all optimization goals without requiring arbitrary weighting factors, leading to more balanced design solutions.